

Problem Set 2 – Suggested Answers

Exercises

1. BLP with two regressors via calculus.

Define the objective function

$$Q(b_2, b_3) := E((Y - X_2b_2 - X_3b_3)^2).$$

Expanding the square inside the expectation and using linearity of $E(\cdot)$:

$$Q(b_2, b_3) = E(Y^2) - 2b_2E(X_2Y) - 2b_3E(X_3Y) + b_2^2E(X_2^2) + 2b_2b_3E(X_2X_3) + b_3^2E(X_3^2).$$

Introduce shorthand notation: $m_{22} := E(X_2^2)$, $m_{33} := E(X_3^2)$, $m_{23} := E(X_2X_3)$, $m_{2Y} := E(X_2Y)$, $m_{3Y} := E(X_3Y)$.

First-order conditions. Differentiating with respect to b_2 and b_3 and setting the partial derivatives to zero:

$$\begin{aligned} \frac{\partial Q}{\partial b_2} &= -2m_{2Y} + 2m_{22}b_2 + 2m_{23}b_3 = 0, \\ \frac{\partial Q}{\partial b_3} &= -2m_{3Y} + 2m_{23}b_2 + 2m_{33}b_3 = 0. \end{aligned}$$

Dividing through by two yields the *normal equations* (a 2×2 linear system in the unknowns b_2, b_3):

$$m_{22}b_2 + m_{23}b_3 = m_{2Y}, \tag{1}$$

$$m_{23}b_2 + m_{33}b_3 = m_{3Y}. \tag{2}$$

Second-order conditions. The Hessian of Q is $\begin{pmatrix} 2m_{22} & 2m_{23} \\ 2m_{23} & 2m_{33} \end{pmatrix}$, which equals $2E\begin{pmatrix} X_2 \\ X_3 \end{pmatrix}\begin{pmatrix} X_2 & X_3 \end{pmatrix}$ and is positive semi-definite by construction. Q is therefore convex in (b_2, b_3) , so any solution of the FOCs is a global minimum.

Solving the normal equations. Multiply (1) by m_{33} and (2) by m_{23} , then subtract:

$$(m_{22}m_{33} - m_{23}^2)b_2 = m_{33}m_{2Y} - m_{23}m_{3Y}.$$

Similarly, multiply (1) by m_{23} and (2) by m_{22} , then subtract:

$$(m_{22}m_{33} - m_{23}^2)b_3 = m_{22}m_{3Y} - m_{23}m_{2Y}.$$

Provided the denominator $m_{22}m_{33} - m_{23}^2 \neq 0$ (i.e. X_2 and X_3 are not perfectly collinear), the unique solution is

$$\begin{aligned}\tilde{\beta}_2 &= \frac{m_{33}m_{2Y} - m_{23}m_{3Y}}{m_{22}m_{33} - m_{23}^2}, \\ \tilde{\beta}_3 &= \frac{m_{22}m_{3Y} - m_{23}m_{2Y}}{m_{22}m_{33} - m_{23}^2}.\end{aligned}$$

same as Exercise 7
in P51

Comparison.

- *Projection on $\text{sp}(X_2, X_3)$ (Assignment 1, Exercise 7):* These are *exactly* the same formulas. This is not a coincidence—both approaches characterise the best linear predictor of Y in $\text{sp}(X_2, X_3)$. The projection approach uses the geometry of Hilbert spaces (orthonormal bases, Gram–Schmidt), while the calculus approach minimises squared prediction error directly. They are two routes to the same destination.
- *Projection on $\text{sp}(1, X_2, X_3)$ (lecture):* Adding a constant shifts the problem. The FOC with respect to b_0 delivers $E(Y) = b_0 + E(X_2)b_2 + E(X_3)b_3$, which can be used to eliminate b_0 . The resulting formulas for b_2 and b_3 look identical in structure, but with *centred* moments (covariances, variances) replacing the raw second moments m_{jk} . In other words, demeaning the variables first reduces the problem with a constant to the problem without a constant.

2. Mean and variance of the sample average.

Expected value. By linearity of $E(\cdot)$:

$$E\bar{Y}_N = E\left(\frac{1}{N} \sum_{i=1}^N Y_i\right) = \frac{1}{N} \sum_{i=1}^N EY_i = \frac{1}{N} \cdot N\mu_Y = \mu_Y.$$

The sample average is an unbiased estimator of μ_Y .

Variance. Using independence of the Y_i (so all cross-covariance terms vanish) and identical distributions:

$$\text{Var}\bar{Y}_N = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N Y_i\right) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}Y_i = \frac{1}{N^2} \cdot N\sigma_Y^2 = \frac{\sigma_Y^2}{N}.$$

Key point: independence is essential here. Without it, the cross-terms $\text{Cov}(Y_i, Y_j)$ for $i \neq j$ would not vanish and the formula would be more complicated. The $1/N$ decay in variance is the engine behind the law of large numbers.

3. Variance of the standardised sample average.

Since $E\bar{Y}_N$ is a constant (a non-random quantity), shifting by a constant does not affect the variance:

$$\text{Var}Z_N = \text{Var}\left(\frac{\bar{Y}_N - E\bar{Y}_N}{\sqrt{\text{Var}\bar{Y}_N}}\right) = \frac{\text{Var}(\bar{Y}_N - E\bar{Y}_N)}{\text{Var}\bar{Y}_N} = \frac{\text{Var}\bar{Y}_N}{\text{Var}\bar{Y}_N} = 1.$$

Why this matters for the CLT. A sequence of distributions that converges to a point mass (degenerate distribution) has variance tending to zero. Since $\text{Var}Z_N = 1$ for all N , the sequence $\{Z_N\}$ cannot degenerate—it keeps spread of order one. This is a *necessary* (though not sufficient) condition for the limit distribution to be non-degenerate, consistent with the CLT conclusion that $Z_N \xrightarrow{d} N(0, 1)$.

4. OLS estimator via matrix calculus.

Expand the objective function:

$$\begin{aligned} Q(b) &:= (Y - Xb)'(Y - Xb) = Y'Y - b'X'Y - Y'Xb + b'X'Xb \\ &= Y'Y - 2b'X'Y + b'X'Xb, \end{aligned}$$

where we used the fact that $b'X'Y = (b'X'Y)' = Y'Xb$ (a scalar equals its own transpose).

First-order condition. Applying the matrix calculus lemma with $z = b$ and $A = X'Y$ for the linear term, and $A = X'X$ for the quadratic term:

$$\frac{\partial Q}{\partial b} = -2X'Y + (X'X + (X'X)')b = -2X'Y + 2X'Xb,$$

where the last equality uses the fact that $X'X$ is symmetric ($(X'X)' = X'X$). Setting the gradient to zero:

$$X'Xb = X'Y.$$

These are the matrix form of the *normal equations*. If $X'X$ is invertible (i.e. X has full column rank K), the unique solution is

$$\hat{\beta}^{\text{OLS}} = (X'X)^{-1}X'Y.$$

Second-order condition. The Hessian is $\partial^2 Q / \partial b \partial b' = 2X'X$, which is positive semi-definite (positive definite when X has full column rank). Hence the FOC yields a global minimum.

Remark. Invertibility of $X'X$ requires $\text{rank}(X) = K$ —no perfect multicollinearity. This is an assumption about the data, not an assumption we can always verify before estimating.

5. A deterministic sequence is $o_p(1)$.

We need to show that $\lambda/N \xrightarrow{p} 0$ as $N \rightarrow \infty$. By definition, this requires

$$\lim_{N \rightarrow \infty} \Pr\left(\left|\frac{\lambda}{N}\right| > \varepsilon\right) = 0 \quad \text{for every } \varepsilon > 0.$$

Since λ is a fixed real number, $|\lambda/N|$ is a *deterministic* sequence converging to zero. Formally, given any $\varepsilon > 0$, choose $N^* > |\lambda|/\varepsilon$. Then for all $N \geq N^*$:

$$\left|\frac{\lambda}{N}\right| \leq \frac{|\lambda|}{N^*} < \varepsilon.$$

Because the event $\{|\lambda/N| > \varepsilon\}$ is *impossible* for $N \geq N^*$, its probability is zero:

$$\Pr\left(\left|\frac{\lambda}{N}\right| > \varepsilon\right) = 0 \quad \text{for all } N \geq N^*.$$

Hence the limit is zero, confirming $\lambda/N = o_p(1)$.

Lesson. Deterministic convergence to zero ($\lambda/N \rightarrow 0$) implies convergence in probability to zero, but not vice versa. The $o_p(1)$ notation is strictly more general; it encompasses both deterministic and stochastic sequences. This exercise motivates why we can freely treat λ/N as $o_p(1)$ in Exercise 6.

6. Consistency of the Ridge regression estimator.

This estimator is known as the *Ridge regression estimator*. The penalty term λI_K prevents $X'X$ from being singular, which is useful when K is large relative to N or when there is near-collinearity. The question is whether this bias-inducing regularisation disappears asymptotically.

Step 1: Substitute the DGP. Replace $Y_i = X_i' \beta^* + u_i$:

$$\begin{aligned} \hat{\theta} &= \left(\sum_{i=1}^N X_i X_i' + \lambda I_K \right)^{-1} \sum_{i=1}^N X_i (X_i' \beta^* + u_i) \\ &= \left(\sum_{i=1}^N X_i X_i' + \lambda I_K \right)^{-1} \left(\sum_{i=1}^N X_i X_i' \right) \beta^* + \left(\sum_{i=1}^N X_i X_i' + \lambda I_K \right)^{-1} \sum_{i=1}^N X_i u_i. \end{aligned}$$

Step 2: Divide by N . Dividing numerator and denominator by N (valid since $N > 0$):

$$\hat{\theta} = \underbrace{\left(\frac{1}{N} \sum_{i=1}^N X_i X_i' + \frac{\lambda}{N} I_K \right)^{-1}}_{=: A_N^{-1}} \underbrace{\frac{1}{N} \sum_{i=1}^N X_i X_i' \beta^*}_{=: B_N} + A_N^{-1} \underbrace{\frac{1}{N} \sum_{i=1}^N X_i u_i}_{=: C_N}. \quad (3)$$

Step 3: Apply the WLLN. Under the maintained assumptions (X_i, Y_i i.i.d., finite second moments), the Weak Law of Large Numbers gives:

$$\begin{aligned} B_N &= \frac{1}{N} \sum_{i=1}^N X_i X_i' = E(X_i X_i') + o_p(1), \\ C_N &= \frac{1}{N} \sum_{i=1}^N X_i u_i = E(X_i u_i) + o_p(1) = 0 + o_p(1) = o_p(1), \end{aligned}$$

where the second line uses the assumption $E(X_i u_i) = 0$.

Step 4: Handle the λ/N term (using Exercise 5). By Exercise 5, $\lambda I_K / N = o_p(1)$. Therefore:

$$A_N = \frac{1}{N} \sum_{i=1}^N X_i X_i' + \frac{\lambda}{N} I_K = E(X_i X_i') + o_p(1) + o_p(1) = E(X_i X_i') + o_p(1).$$

Step 5: Invert (Slutsky's theorem / CMT). Assuming $E(X_i X_i')$ is positive definite (and hence invertible), matrix inversion is a continuous operation at $E(X_i X_i')$. By the continuous mapping theorem:

$$A_N^{-1} = (E(X_i X_i') + o_p(1))^{-1} = E(X_i X_i')^{-1} + o_p(1).$$

Denote $M := E(X_i X_i')^{-1}$ (a fixed $K \times K$ matrix, hence $O_p(1)$).

Step 6: Combine using $O_p(1) \cdot o_p(1) = o_p(1)$. Substituting back into (3) and expanding step by step:

$$\begin{aligned} \hat{\theta} &= (M + o_p(1))(M^{-1} + o_p(1))\beta^* + (M + o_p(1))o_p(1) \\ &= (I_K + M o_p(1) + o_p(1)M^{-1} + o_p(1)o_p(1))\beta^* + O_p(1) \cdot o_p(1) \\ &= (I_K + o_p(1) + o_p(1) + o_p(1))\beta^* + o_p(1) \\ &= (I_K + o_p(1))\beta^* + o_p(1) \\ &= \beta^* + \beta^* o_p(1) + o_p(1) \\ &= \beta^* + o_p(1) + o_p(1) \\ &= \beta^* + o_p(1). \end{aligned}$$

Here we used $O_p(1) \cdot o_p(1) = o_p(1)$ at the line with $(M + o_p(1))o_p(1)$, and the fact that β^* is a fixed vector (hence $O_p(1)$).

Conclusion. Since $\hat{\theta} = \beta^* + o_p(1)$, we have $\hat{\theta} \xrightarrow{P} \beta^*$. The Ridge regression estimator is *consistent* for β^* .

Intuition. The Ridge penalty λI_K shrinks the estimator toward zero, introducing bias in finite samples. However, the penalty is fixed while the sample size grows, so $\lambda/N \rightarrow 0$: the relative weight of the penalty vanishes asymptotically. This is why consistency is recovered even though Ridge is biased in finite samples.

$$\hat{\theta} := \left(\left(\sum_{i=1}^N X_i X_i' \right) + \lambda I_K \right)^{-1} \left(\sum_{i=1}^N X_i Y_i \right),$$

$$= \left(\left(\sum_{i=1}^N X_i X_i' \right) + \lambda I_K \right)^{-1} \left(\sum X_i (X_i' \beta^* + u_i) \right)$$

$$= \left(\left(\sum_{i=1}^N X_i X_i' \right) + \lambda I_K \right)^{-1} \left(\sum X_i X_i' \right) \beta^*$$

$$+ \left(\left(\sum_{i=1}^N X_i X_i' \right) + \lambda I_K \right)^{-1} \left(\sum X_i u_i \right)$$

$$= \left[\left(\frac{1}{N} \sum X_i X_i' \right) + \frac{\lambda}{N} I_K \right]^{-1} \left(\frac{1}{N} \sum X_i X_i' \right) \beta^*$$

$$+ \underbrace{\left[\left(\frac{1}{N} \sum X_i X_i' \right) + \frac{\lambda}{N} I_K \right]^{-1}}_{A_N} \underbrace{\left(\frac{1}{N} \sum X_i u_i \right)}_{C_N}$$

It follows: by WLLN

$$A_N = E(X_i X_i') + o_p(1) + o_p(1) = E(X_i X_i') + o_p(1)$$

$$A_N^{-1} = [E(X_i X_i') + o_p(1)]^{-1} = E(X_i X_i')^{-1} + o_p(1) \quad \text{by Slutsky Thm}$$

assume that $E(X_i X_i')^{-1}$ exists

$$B_N = E(X_i X_i') + o_p(1)$$

$$C_N = E(X_i u_i) + o_p(1) = 0 + o_p(1) = o_p(1)$$

Therefore

$$\begin{aligned}\hat{\theta} &= A_N^{-1} B_N \beta^* + A_N^{-1} C_N \\&= \left[E(X_i X_i')^{-1} + o_p(1) \right] \left[E(X_i X_i') + o_p(1) \right] \beta^* \\&\quad + \left[E(X_i X_i')^{-1} + o_p(1) \right] \cdot o_p(1) \\&= \left[I_K + E(X_i X_i')^{-1} \cdot o_p(1) + o_p(1) \cdot E(X_i X_i') + o_p(1) \cdot o_p(1) \right] \beta^* \\&\quad + E(X_i X_i')^{-1} \cdot o_p(1) + o_p(1) \cdot o_p(1) \\&= \left[I_K + O_p(1) \cdot o_p(1) + o_p(1) \cdot O_p(1) + o_p(1) \right] \beta^* \\&\quad + O_p(1) \cdot o_p(1) + o_p(1) \\&= \left[I_K + o_p(1) + o_p(1) + o_p(1) \right] \beta^* \\&\quad + o_p(1) + o_p(1) \\&= \left[I_K + o_p(1) \right] \beta^* + o_p(1) \\&= \beta^* + o_p(1) \cdot \beta^* + o_p(1)\end{aligned}$$

$$= \beta^* + o_p(1) + o_p(1)$$

$$= \beta^* + o_p(1)$$

$$\underline{\underline{\quad\quad\quad}}$$